

CURRENT FILE

Bt401

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1. Germplasm 6

Germplasm has to be maintained in such a state that minimizes the risk of loss and allows either is direct planting in the field or its preparation of planting with relative ease this is called plant genetic resources conservation or germplasm conservation

2. Policy and strategy 11

Policy: A policy is a set of basic principles and guidelines, formulated and enforced by national government to direct and limit its action in pursuit of long-term goals

Strategy: A strategy refers to a concrete plan of action, specifically designed to achieve concrete goals

3. Name of plant of nut 13-14

- ♥ Almonds
- ♥ Walnut
- ♥ Tree Nuts

4. Threats to AnGR 15

- ♥ The diversity of animal genetic resources has decreased over time despite their importance.
- ♥ A threat to livestock diversity is pressure from large-scale commercial production systems to maintain only high-output breeds.
- ♥ Changes in climate will have an impact on livestock and food production
- ♥ Some major disease threats livestock include rinderpest, foot and mouth disease, and Peste des petits ruminants (PPR), also known as sheep and goat plague.

5. Genetic drift 20-21

Genetic drift is process of change in the frequency of alleles in a population of organisms due to random chance

6. Define endangered species with examples 20-21

“A species of plant or animal that is in immediate danger of becoming extinct and needs protection to survive.” e.g. Gray Wolves

7. Genetic diversity effect on environment give example 24

The huge variety of different gene sets define an individual or a whole population's ability to tolerate stress from any given environmental factor. Some individuals might be able to tolerate an increased load of pollutants in their environment. Others carrying different genes might suffer from infertility or even die under the exact same environmental conditions.

Example: In general, woody plants like trees have a higher genetic diversity than vascular plants like grasses.

8. Causes of gene pool 31-32

The allele frequency in gene pool can be affected by evolutionary mechanisms, such as; Mutation, Gene flow, Inbreeding, Natural selection, Founder effect and Random genetic drift

9. Purpose of grading up 41

Breeding of animals of two different breeds where the animals of an indigenous breed/genetic group are mated by an improved pure breed for several generations towards attaining the superior traits of the improved breed. Grading up is continuous use of purebred sires of the same breed in a grade herd. By fifth generation, the graded animals may reach almost purebred levels.

10.Cold storage 59

Cold storage: it involves storage in low and non-freezing temperature.

Slow Freezing- Slow thawing: With this method organs are labeled into vials after equilibration with a Cryoprotectant solution and then cooled at rate of 0.5-2 C per minute down to -1 C. seeding is then induced and a holding period of 5 to 15 minutes allows equilibration of the temperature. Thereafter embryos are cooled to -60 C or lower at a rate of 0.3 to 0.5 C per minute before being transfer to liquid nitrogen. Frozen embryos must be slowly thawed at a rate of less than 25 C per minute to prevent osmotic shock

Rapid cooling and rapid thawing: In this technique, however cooling is terminated at -30 to -40 C and embryos are then plunged into liquid nitrogen for rapid cooling to -196 C. Thawing is therefore performed rapidly (200 to 500 C per minute) to prevent recrystallization

11.Role of cryoprotectants in cryopreservation 60

A cryoprotectant is a substance used to protect biological tissue from freezing damage. Due to ice crystal formation cell membrane and cell integrity can ruptured. cryoprotectants are usually antifreeze compound.

12.Cryo conversation of animals genetic resources 61

Cryo-conservation of animal genetic resources is a strategy wherein samples of animal genetic materials are preserved cryogenically.

13.Freezing method 61

Cryopreservation is based on the ability of certain small molecules to enter cells and prevent dehydration and formation of intracellular ice crystals, which can cause cell death and destruction of cell organelles during the freezing process. The sensitivity of cells to low temperature depends on the plant species. There are four different types of methods:

- ♥ **Slow freezing method:** the tissue or plant material is slowly frozen at slow cooling rate. The advantage is the plant cells are partially dehydrated and survive better.
- ♥ **Rapid freezing method:** it involves plunging the vials in liquid nitrogen. The temperature decreases from -300 to -1000 degree rapidly.
- ♥ **Combined freezing method:** this is combination of both slow and rapid freezing method. The process is carried out in step wise like manner.
- ♥ **Dry freezing method:** in this method dehydrated cells and seeds are stored

14.Role of National Park in Landscape and Seascape? 67

- ♥ Protecting larger-scale ecological processes
- ♥ Protecting compatible ecosystem services
- ♥ Protecting particular species and communities that require relatively large areas of undisturbed habitat
- ♥ To inform and excite visitors about the need for and potential of conservation programmes
- ♥ To support compatible economic development, mostly through recreation and tourism, that can contribute to local and national economies and in particular to local communities.

15.Socio culture values of national monument features 68-69

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. These areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism.

16.Category III includes laws 68-69

The classification of a Category III protected area requires that such areas are managed for conservation by legal or other effective means, and therefore legal recognition and protection at the national or subnational level is likely to be present in these sites. The level of legal protection will however vary between countries, and will depend on the governance type of the area, as they receive differing levels of recognition by government in different countries. Nonetheless a number of national laws are likely to apply to these sites that deter large-scale economic activities in order to maintain the conservation values of these areas

17.Features of management area 70-71

- ♥ Protection of particular species
- ♥ Protection of habitats
- ♥ Active management to maintain target species
- ♥ Active management of culturally-defined ecosystems

18.Primary objective of IV category? 70-71

- ♥ **Primary objective:** To maintain, conserve and restore species and habitats.
- ♥ **Other objectives:** • To protect vegetation patterns. • To protect fragments of habitats as components of landscape or seascape-scale conservation strategies. • To develop public education and appreciation of the species or habitat • Developed by the International Union for Conservation of Nature (IUCN) with support of the World Commission on Protected Areas (WCPA) and other international institutions such as the Convention on Biological Diversity (CBD), and assigned by national governments.

19.Category V Consideration 72-73

- ♥ To maintain a balanced interaction of nature and culture.
- ♥ To contribute to broad-scale conservation by maintaining species associated with cultural
- ♥ landscapes and/or by providing conservation opportunities in heavily used landscapes;
- ♥ To provide opportunities for enjoyment, well-being and socio-economic activity through
- ♥ recreation and tourism;
- ♥ To provide natural products and environmental services;
- ♥ To provide a framework to underpin active involvement by the community in the management
- ♥ of valued landscapes or seascapes and the natural and cultural heritage that they contain;
- ♥ To encourage the conservation of agrobiodiversity and aquatic biodiversity;
- ♥ To act as models of sustainability so that lessons can be learnt for wider application

20.Objective of Category VI 74-75

- ♥ To protect natural ecosystems

- ♥ Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial

21. Three botanical garden 80-81

- ♥ Cacti and succulent plants
- ♥ Herb gardens
- ♥ Greenhouses, shade houses
- ♥ Tropical plants
- ♥ Medicinal Plants
- ♥ Aromatic or textile plants
- ♥ Other exotic plants.

22. Who is responsible for botanical garden 80-81

Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science.

23. What are the role of conservation of botanical Garden 80-81

Conservation of rare and threatened plants. The conservation of plant diversity is critical for sustainable development and botanic gardens are playing a key role as centers of conservation action. Botanical gardens can promote diversity. Because they include many species of plant

24. Explain the history of Ramsar convention? 82-84

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975

25. Three Pillers of Ramsar Site? 82-84

- ♥ Work towards the wise use of all their wetlands.
- ♥ Designate suitable wetlands for the list of Wetlands of International Importance.
- ♥ Cooperate internationally on transboundary wetlands, shared wetland systems and shared species.

26. List of Ramsar 82-84

Baluchistan: Astola Island • Hub Dam • Jiwani Coastal Wetland • Miani Hor • Ormara Turtle beach

Sindh: Keenjhar (Kalri) Lake • Haleji Lake • Drigh Lake • Indus Dolphin reserve • Jubho lagoon • Nurri Lagoon • Deh Akro-II • Indus Delta • Runn of Kutch

Punjab: Uchhali Complex • Taunsa Barrage • Chashma Barrage

Khyber Pakhtunkhwa: Tanda Dam • Thanedar Wala

27. Name of Ramsar in Punjab? 82-84

- ♥ Uchhali Complex
- ♥ Taunsa Barrage
- ♥ Chashma Barrage

28. Three pillars of Ramsar convention 82-84

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- ♥ Designate suitable wetlands for the list of Wetlands of International Importance.
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29. Define game reserve? 87

“A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis”

30. Need of genetic preservation 89

Conservation of genetic resources is essential for the resilience and stability of populations and the persistence of species, and is a means of safeguarding living materials exploited by agriculture, industry, forestry, and aquaculture.

22. Vivo gene bank? 89

Generally, plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank.

31. Physical growth limitation by storage 91

- ♥ Low temperature
- ♥ Low light/restricted photoperiod
- ♥ Minimal containment
- ♥ Minimal O₂
- ♥ Osmotic (water) stress

32. Tissue conservation 91

Conservation in tissue culture in in vitro genebanks is often combined with cryopreservation. Cultures in the active genebank are maintained by successive subculturing allowing culture renewal and distribution. For medium term storage, sub-culture intervals are extended, reducing processing costs by arresting growth using cold treatments, adapted light conditions, culture medium modifications (osmotic active compounds, growth retardants). This increases efficient use of resources and staff time and offsets selection risks and contamination

33. DNA banks 92

DNA banks can now be considered as a means of complimentary conservation. DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field Gene banks and nor conserved in situ due to high risk in that area.

Advantages: DNA banking is an efficient, simple and long-term method to conserve the genetic information.

Disadvantages: There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

34. What are the strategy of DNA bank 92

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Long-term needs and expected volume and number of samples to be stored will determine organization and repository design.

35. Which tissue are used in static natural 93-95

The ex-situ conservation (static) strategies include botanical gardens, zoological gardens, conservation stands and gene, pollen, seed, seedling, tissue culture and DNA banks. Seed gene banks make the easiest way to store germplasm of wild and cultivated plants at low temperature.

36. Limitation of germplasm conservation 96

- ♥ Seeds of many important tree species cannot adequately survive long-term storage;
- ♥ Seeds of many tropical hardwoods cannot survive even a year of conventional seed storage;

- ♥ Some genetic damage or change in gene frequencies may occur during seed storage. Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required. Some advantages of cryogenic storage are:

37.Effect of climate change on invertebrates' genetic resources? 98

Climate change is expected to affect all three of the main groups of invertebrate ecosystem-service providers as well as invertebrate pests. Invertebrates have limited ability to control their body temperatures. Therefore, although some groups such as soil-dwelling organisms are to some degree buffered against the effects of temperature fluctuations in the wider environment, it is likely that rising temperatures will directly influence the distribution of invertebrate species. Many of the challenges associated with the management of invertebrate genetic resources in agriculture in the context of climate change will relate to climate-driven or human-assisted movement of invertebrate species.

38.Human activities in invertebrate genetic resources 98

Human activities have created barriers to the migration of invertebrate species. These barriers are likely to affect species in natural ecosystems rather more severely than those associated with agro-ecosystems. The movement of the latter is likely to be facilitated rather than hindered by human-induced landscape changes. In situ adaptation of invertebrate species is expected to be most marked where movement is not an option

39.Role of pollinators 100

Nearly ninety percent of wild flowering plants need pollinators like bees to transfer pollen for successful sexual reproduction .in turn, these plants are critical in the functioning of ecosystems. They provide food, form habitats and provide a wide range of other resources for many animal species.

40.Forest genetic resources 101

Forest genetic resources or tree genetic resources are genetic material of shrub and tree species of actual or future value. Forest denotes a stand, population or landscape of trees, and typically other associated woody plants. Definition: Forest genetic resources are essential for forest-dependent communities who rely for a substantial part of their livelihoods on timber and non-timber forest products (for example fruits, gums and resins) for food security, domestic use and income generation.

41.Write the Variations In Variation Gene 101

- ♥ Variation between species,
- ♥ Variation between populations within species
- ♥ Variation between individual trees within populations. The largest variation is between species, and loss of whole species is therefore also the most dramatic loss of future options.

42.The role of phenotypic plasticity 104

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions.

43.Law and legislations 108-109

Law: The system of rules which a particular country or community recognizes as regulating the actions of its members and which it may enforce by the impossible penalties.

Legislations: The process of making or enacting laws.

44. Maritime Zones act 1976 108-109

- ♥ It includes provisions for preservation, development and protection of marine environment
- ♥ This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- ♥ This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

45. Historical Background of CBD? 110

- ♥ CBD is about the conservation and wise use of different biological resources (plants and animals).
- ♥ It was adopted in 1992 at Rio De Janeiro, Brazil and entered force on January, 1993, which was 90 days after the 30th ratification
- ♥ Pakistan signed it in June 1992 at United Nations Conference on Environment and Development held at Rio De Janeiro, Brazil

46. Objective of CBD 110

The conservation of biological diversity:

- ♥ The sustainable use of its components;
- ♥ The fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate
- ♥ Access to genetic resources,
- ♥ Transfer of relevant technologies,
- ♥ Funding.

47. Area of Nagoya protocol 112

The Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity, also known as the Nagoya Protocol on Access and Benefit Sharing (ABS) is a 2010 supplementary agreement to the 1992 Convention on Biological Diversity (CBD).

48. Cartagena protocol 113

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009

Objectives of Cartagena Protocol on Biosafety

- ♥ Contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity,
- ♥ It takes into account risks to human health, and specifically focusing on transboundary movements.
- ♥ It seeks to protect biodiversity from the potential risks of living modified organisms (LMOs) resulting from modern biotechnology.

49. Measures for importing plants and their materials 116

All persons who intend to export plant materials must submit an application for inspection of plant material before the dispatch of consignments. If the plant or plant material is found to be free from plant diseases and injurious insects, a certificate must be issued by the Director or Entomologist (Quarantine) to the exporter to accompany the shipment. If the Director requires

the disinfestations or disinfection of plants or plant materials by fumigation, the Department may register any persons, firm, agency organization or company. No official certificate shall be granted for plants or plant material which have been from or mixed with other plants which are diseased or infested, and any risk or damages resulting from fumigation or other treatment must be at the risk of the owner.

50.Strategies of effective conservation of AnGR 119

Following strategies should be followed for effective conservation and utilization of AnGR:

- ♥ Formulating the National Livestock and wild-life Breeding Policies
- ♥ Encouraging the Formation of Breed Associations
- ♥ Developing Professional Human Resources
- ♥ Strengthening Research and Development Institutions
- ♥ Developing Infrastructure for marketing International co-operation and assistance is needed in capacity building to remodel available livestock farms/research stations to conserve and develop genetic resources

51.Angr limitation 119

Much work is being done for the conservation of plant genetic resources but animal genetic resources (AnGR) still need attention.

Following strategies should be followed for effective conservation and utilization of AnGR:

- ♥ Formulating the National Livestock and wild-life Breeding Policies.
- ♥ Encouraging the Formation of Breed Associations.
- ♥ Developing Professional Human Resources.
- ♥ Strengthening Research and Development Institutions
- ♥ Developing Infrastructure for marketing International co-operation and assistance is needed in capacity building to remodel available livestock farms/research stations to conserve and develop genetic resources.

52.Which area are cover under Cartagena Protocol 113

Transboundary movement, transit, handling and use of all living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity, taking into account risks to human health.

53.*Cold storage